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PERFORMANCE SPECIFICATION

PROPELLANT, HYDRAZINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

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1. SCOPE

1.1 Scope. This specification covers the requirements for two grades of hydrazine (N_2H_4).

1.2 Classification. Hydrazine will be of the following designated grades as specified (see 6.2).

GRADES

Monopropellant (MPH)	-	Normal production.
High Purity (HPH)	-	Special production with strict control of specified impurities.

1.3 Part or identifying number (PIN). The PINs to be used for hydrazine acquired to this specification are created as follows:

M 26536 – X Example of PIN: M26536-S

- X: Grade (see 1.2). Use “M” for monopropellant; “H” for high purity.
- 26536: Specification number.
- M: Prefix for military specification.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-25604	Propellant, uns-Dimethylhydrazine
MIL-PRF-27401	Propellant Pressurizing Agent, Nitrogen
MIL-PRF-27404	Propellant, Monomethylhydrazine
MIL-PRF-27407	Propellant Pressurizing Agent, Helium

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1068	Standard Test Methods for Iron in Water
ASTM D1976	Standard Test Method for Elements in Water by Inductively-Coupled Plasma Atomic Emission Spectroscopy
ASTM D2276	Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling
ASTM E29	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications
ASTM E394	Standard Test Method for Iron in Trace Quantities Using the 1,10-Phenanthroline Method

(Copies of these documents are online available at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical properties. The chemical and physical properties of the propellant shall conform to those listed in Table I when tested in accordance with (IAW) the applicable test methods.

3.2 Limiting values. The following applies to all specified limits in this specification. For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off “to the nearest unit” in the last right-hand digit used in expressing the specification limit according to the Rounding Method of ASTM E29.

3.3 Filter. A filter with a rating of at least 2 micrometer (µm) nominal and 10 µm absolute rating shall be installed between the manufacturer’s plant system and the container to be filled for delivery.

3.4 Qualitative. The propellant shall be colorless, homogeneous liquid when examined visually by transmitted light.

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TABLE I. *Chemical and physical properties.*

Properties	Monopropellant Grade (MPH)	High Purity Grade (HPH)	Test Paragraph
Hydrazine wt%, min.	98.5	99.0	4.3.2
Water wt%, max.	1.0	0.75	4.3.2
Ammonia wt%, max.	---	0.3	4.3.2
Particulate mg/L, max.	1.0	1.0	4.3.3
Chloride mg/kg (ppmw), max.	5	5	4.3.4
Iron mg/kg (ppmw), max.	20	1.0	4.3.5
Non-volatile residue mg/L, max.	50	10	4.3.6
Aniline wt%, max	0.50	0.003	4.3.7
Carbon dioxide(1) wt%, max	0.003	0.003	4.3.8
Carbonaceous Material, (Total of OVCM(2) + OCM(3)), wt%, max	0.02	0.005	4.3.9

NOTES:

(1) Determination of carbon dioxide (CO_2) content must be performed before all other analyses to prevent further absorption of CO_2.

(2) OVCM – Other volatile carbonaceous material as the total of unsymmetrical dimethylhydrazine (UDMH), isopropyl alcohol (IPA), and monomethylhydrazine (MMH),

(3) OCM – Other carbonaceous material consisting of any impurity providing a positive FID response ahead of and trailing the hydrazine peak excluding OVCM and aniline.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as conformance inspections.

4.2 Conformance inspection. The conformance inspection shall consist of the following:

a. Individual tests (see 4.2.1).

b. Sampling tests (see 4.2.2).

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4.2.1 Individual tests. The propellant shall be subjected to the following test as described under 4.3:

a. Examination of product (see 4.3.1).

4.2.2 Sampling tests. The propellant shall be selected according to 4.2.2.1 and subjected to the tests in Table I as described under 4.3.

4.2.2.1 Sampling plan. Unless otherwise specified (6.2), each filled shipping container shall be considered a lot and shall be sampled.

4.2.2.1.1 Sample. A sample consists of not less than 1500 milliliters (mL) of propellant for all tests specified in 4.2.2. Unless otherwise specified, conformance tests shall be made on the sample of propellant taken directly from the shipping container. When required, the sample shall be forwarded to a laboratory designated by the procuring activity for subjection to the conformance tests specified herein. The clear glass narrow mouth bottles intended for sampling shall be specially cleaned and handled according to the procedure described in ASTM D2276 Section 10 (Preparation of Flushing Apparatus) with the exception listed in 4.3.3d. Sampling shall be conducted in such a manner that atmospheric exposure of the contents of the shipping container and the sample is minimized.

4.2.3 Rejection. When any sample of the propellant tested IAW 4.3 fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected.

4.3 Test methods.

4.3.1 Examination of product. The propellant shall be visually examined to ensure that the material conforms to 3.4. The examination shall be conducted after a representative aliquot of the sample has been transferred to a clear glass vessel and compared to the same type vessel containing the same volume of purified water.

4.3.2 Hydrazine assay and water. The hydrazine (N_2H_4) and water (H_2O) content of the propellant shall be determined by one of the gas chromatographic (GC) methods described in Appendix A. In case of a dispute, the capillary column method (A.2.1) shall be the referee.

4.3.3 Particulate. The propellant sample shall be tested gravimetrically for particulate contamination in accordance with ASTM D2276, with the following exceptions:

a. Mix the sample thoroughly without exposure to air. Immediately pour no less than 250 mL of the sample into a clean graduated cylinder.

b. Pass the sample aliquot through a Mitex disc polytetrafluoroethylene (PTFE) hydrophobic, 5 µm, 47 mm, white filter or equivalent.

c. The oven temperature shall be set at 70 °C (158 °F).

d. Filtered American Chemical Society (ACS) reagent grade isopropyl alcohol shall be used for rinsing the sample bottle and filter holder instead of petroleum ether specified in ASTM D2276. Use a suitable solvent resistant filter disc with an absolute rating of at least 1.0 µm. Wet the filter with the filtered isopropyl alcohol prior to filtering the sample.

4.3.4 Chloride. The chloride (Cl^-) content of the propellant shall be determined by one of the methods described in Appendix B. In case of a dispute, the specific ion electrode method (B.2.1) shall be the referee.

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4.3.5 Iron. The iron (Fe), content of the propellant shall be determined by one of the methods described in Appendix C. In case of a dispute, the inductively-coupled plasma method (C.2.1) shall be the referee.

4.3.6 Non-volatile residue. The non-volatile residue (NVR) of the propellant shall be determined in duplicate by one of the methods in Appendix D. In case of a dispute, the hot plate method (D.2.1) shall be the referee.

4.3.7 Aniline. The aniline ($C_6H_5NH_2$) content of the propellant shall be determined by one of the methods found in Appendix E. In case of a dispute the spectrophotometric method (E.2.1) shall be the referee method for monopropellant and the gas chromatographic method (E.2.2) for high purity.

4.3.8 Carbon dioxide. The carbon dioxide (CO_2) content of the propellant shall be determined by the method described in Appendix F.

4.3.9 Carbonaceous material. Other volatile carbonaceous material (OVCM) + other carbonaceous material (OCM) shall be determined by the gas chromatographic method in Appendix G.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. The propellant described by this specification is intended for use as a fuel in rocket engines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Grade of the propellant required (1.2).

c. Method of shipment, type and capacity of containers.

d. Quantity by weight in pounds (avoirdupois).

e. When a different sampling plan is required (4.2.2.1).

f. When test methods are other than those specified (4.3).

g. Packaging requirements (5.1).

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6.3 Safety Data Sheets. Contracting officers will identify those activities that require copies of completed Safety Data Sheets (SDS) prepared in accordance with the 29CFR1910.1200.

6.4 Subject term (key word) listing.

Catalytic Engine

Fuel

Monopropellant

Rocket Engine

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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APPENDIX A

HYDRAZINE ASSAY AND WATER

A.1 SCOPE

A.1.1 Scope. This Appendix describes two gas chromatographic (GC) methods for the determination of hydrazine and water content in the sample. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 METHODS

A.2.1 Gas chromatographic method - capillary column.

A.2.1.1 Procedure. A 60 or 30 meters (m) x 0.53 millimeters (mm), 1.00 µm, DB-Wax column, a thermal conductivity detector (TCD), with a split injector and an ultra-inert, deactivated, glass wool packed split liner shall be used. Table A-1 provides a summary of the critical operating parameters. These parameters may vary depending on GC system and detector brands, and may be adjusted by the chemist to provide optimal component resolution.

Allow the instrument to stabilize, condition the column by adding a couple of injections to the sample sequence. Perform check runs by processing three consecutive injections of 1 microliter (µL) of sample. The checks must meet the quality control criteria in A.2.1.3.

This method may be used for aniline quantification for monopropellant hydrazine.

TABLE A-1. *GC parameters.*

PARAMETER	CONDITION
Injector Port Temp	150 °C
Injection Mode	Split injection
Sample Volume.	1 µL
Split Ratio	4:1
Carrier Gas	He
Control Flow	10 mL/min
Control Mode	Flow
TCD Detector Temp.	210 °C
Equilibration Time	2.00 min
Initial Temp.(T_0)	60 °C
Initial Time (t_0)	3.00 min
Rate (1)	10 °C/min
Final Temp. (T_{f1})	100 °C
Final Time (t_{f1})	0.00 min
Rate (2)	20 °C/min
Final Temp. (T_{f2})	185 °C
Final Time (t_{f2})	3.75 min

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A.2.1.2 Calculations.

$$\% N_2H_4 = \frac{A_{N_2H_4} \times 100}{\sum A_i}$$

$$\% H_2O = \frac{A_{H_2O} \times 100}{\sum A_i}$$

$$\% NH_3 = \frac{A_{NH_3} \times 100}{\sum A_i}$$

NOTE: % NH_3 for high purity grade only.

where,

$A_{N_2H_4}$ = The measured area of the N_2H_4 peak multiplied by its signal attenuation factor.

A_{H_2O} = The measured area of the H_2O peak multiplied by its signal attenuation factor.

A_{NH_3} = The measured area of the NH_3 peak multiplied by its signal attenuation factor.

$\sum A_i$ = The sum of all measured peak areas multiplied by their signal attenuation factors.

Assumption: The thermal conductivities of all components in the sample are equal.

A.2.1.3 Quality control. The following performance criteria must be met for fuel purity analysis,

GC replicate analyses (3 consecutive)	RSD[a] ≤ 5 %

a. RSD = Standard Deviation * 100% / Mean

A.2.1.4 Equipment.

(1) Gas chromatograph equipped with a TCD and a split injector.

(2) Agilent ChemStation or equivalent.

(3) A 60 or 30 m x 0.53 mm ID and 1.00 µm film thickness DB-wax capillary column (Agilent P/N 125-7062 or 125-7032 respectively) or equivalent.

(4) Ultra-inert deactivated liner, split, straight, glass wool.

(5) No-bleed septa.

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A.2.2 Gas chromatographic method - packed column.

A.2.2.1 Assay column preparation. Cap one end of a 1/8 inch (in) outer diameter (OD) by 6 foot (ft) nickel tube and fill the tube with the Tenax TA column packing by pouring through a small funnel attached to the other end. Tap or mechanically vibrate the tube to ensure uniform packing. When the tube is filled, plug both ends with a small wad of glass wool, bend the column to the configuration required by the column oven, and connect the column to the inlet fitting in the oven. Condition the column with carrier gas (helium) flowing and the oven set at 200 °C for one hour. After conditioning the column, connect the other end to the detector and set the carrier gas flow to approximately (~) 25 milliliters per minute (mL/min), and the column oven to 50 °C. The inlet and detector temperatures, if separately heated, shall be set to 100 °C and 150 °C, respectively. The detector current should be set to a nominal sensitivity value recommended for helium by the instrument manufacturer. The column temperature and carrier gas flow may be adjusted to provide adequate component resolution for minimum analysis time.

A.2.2.2 Procedure. Equilibrate the column with hydrazine by injection of three or more 1 – 2 µL samples into inlet. If more than 30 minutes (min) elapse between analyses, a single 1 µL injection of hydrazine should re-equilibrate the column. Inject 1 - 2 µL of hydrazine for analysis and record the areas of all peaks in the chromatogram, using the following time/temperature program:

Helium carrier gas flow	20 – 25 mL/min
Initial temperature	50 °C
Rate	20 °C/min
Final temperature	155 °C
Final time hold	4 min

Each analysis should require less than 20 min for elution of all components. The elution order of possible components is as follows:

Air > ammonia> methylamine > water > hydrazine.

A.2.2.3 Calculations. Use equations in A.2.1.2.

A.2.2.4 Equipment and Reagents.

a. Equipment:

(1) Gas chromatograph equipped with a TCD.

(2) Agilent ChemStation or equivalent.

(3) Packed column may be commercially obtained.

b. Reagents:

Tenax TA 60/80 mesh or equivalent.

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CHLORIDE

B.1 SCOPE

B.1.1 Scope. This Appendix describes three methods for the determination of chloride content of the hydrazine. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 METHODS

B.2.1 Specific ion electrode method.

B.2.1.1 Procedure. Rinse the hydrazine NVR in one vessel from 4.3.6 with a 5 mL aliquot of 1N sulfuric acid (H_2SO_4). Briefly stir and heat the liquid to boiling and cover with an inverted watch glass to prevent the condensate from escaping. Transfer the solution to a 25 mL volumetric flask. Rinse vessel with a second 5 mL aliquot of acid followed by one 5 mL aliquot of purified water. Combine the aliquots in the 25 mL volumetric flask and add two to three drops of bromocresol green indicator. Dropwise titrate the contents of the flask to a green endpoint with 1N sodium hydroxide (NaOH) and dilute to the mark with purified water. Mix the solution and transfer ~ 10 mL to a 25 mL beaker. Immerse the electrodes in the sample solution and record the millivolt (mV) reading. Compare the mV reading from the sample to a calibration curve prepared from chloride standards and record the concentration (ppmv) of the sample.

B.2.1.2 Calibration. Prepare an acid matrix blank and a minimum of four calibration standards to bracket the expected chloride (Cl^-) concentration range in the samples by adding one drop of 0.1N H_2SO_4 to 50 mL of purified water in 100 mL volumetric flasks, and combining appropriate aliquots of a 1000 ppm Cl^- standard. Add 2-3 drops of bromocresol green indicator to each of the five flasks and if necessary adjust with 0.1N H_2SO_4 or 0.1N NaOH to obtain a green endpoint. Dilute each to the mark with purified water and mix the solutions thoroughly. Pour enough of the standards into a small beaker to cover the electrode tips and record the resulting mV reading. Prepare a calibration curve of Cl^- concentration to mV and use this to determine Cl^- concentration in the sample.

B.2.1.3 Calculation. Calculate the chloride content of the hydrazine using the following formula:

$$Cl^- ppmw = \frac{(Cl^-_{ppmv})(V_d)}{(V_p)(\rho)}$$

where,

Cl^-_{ppmv} = Chloride content of the sample solution from the calibration curve.

V_d = Final dilution volume taken from NVR in mL.

V_p = Volume of hydrazine used in the NVR analysis in mL.

ρ = Density of hydrazine (1.01 g/mL).

B.2.1.4 Reagents and equipment. The following shall apply as test conditions of B.2.1:

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B.2.1.4.1 Reagents.

a. Purified water free of interfering substances; the water is considered free of interfering substances if a 100 mL sample used to dissolve 0.1 g of silver nitrate, shows no turbidity after standing in the dark for 24 hours. The purified water as described in this paragraph shall be used throughout the procedure.

b. 1N H_2SO_4; prepare by pouring 29 mL of ACS reagent grade concentrated H_2SO_4 into 300 mL of purified water contained in a 1000 mL volumetric flask. Cool the solution to room temperature, dilute to the mark with the purified water, and mix thoroughly. Transfer the solution to a plastic bottle. A 1N H_2SO_4 solution may be commercially obtained.

c. 1N NaOH; prepare by dissolving 40 g of ACS reagent grade, low carbonate, NaOH in 300 mL of purified water contained in a 1000 mL volumetric flask. Cool the solution to room temperature, dilute to the mark with purified water, and mix thoroughly. Transfer the solution to a plastic bottle and protect from long exposure to the atmosphere. A 1N NaOH solution may be commercially obtained.

d. Standard sodium chloride (NaCl) solution, 1 mL = 1.00 mg Cl^-; weigh 1.648 g dry NaCl into a 1 liter (L) volumetric flask. Dilute to the mark with purified water. Transfer the thoroughly mixed solution to a clean polyethylene bottle for storage. A 1000 mg/L (ppmv) chloride standard may be commercially obtained; must be certified.

e. Bromocresol green indicator; 1% aqueous solution. Use purified water.

B.2.1.4.2 Equipment.

a. Chloride electrode – solid state Orion 9417BN, or equivalent.

b. Reference electrode - Orion 900200 Sure-Flow® double junction with 1M potassium nitrate (KNO_3) outer compartment filling solution, or equivalent.

c. pH/voltmeter, Orion 420A+ or equivalent.

B.2.2 Colorimetric method.

B.2.2.1 Procedure. Add to the hydrazine NVR in one vessel from 4.3.6 five milliliters of ferric alum indicator and transfer to a 100 mL volumetric flask. Rinse the vessel with three 5 mL purified water aliquots and combine into the volumetric flask. Set samples aside.

Prepare a calibration blank and at a minimum two standards to bracket the expected Cl^- concentration in the samples in 100 mL volumetric flasks, and add 5 mL of ferric alum indicator. Add 2.5 mL of mercuric thiocyanate solution to each volumetric flask, samples and standards, and dilute to the mark with purified water. Mix the solutions thoroughly and allow to stand in darkness for 15 – 30 min. Analyze in a UV-Vis spectrophotometer at 460 nm, at the wavelength recommended by the instrument manufacturer, or at that which provides the highest sensitivity using 5 cm cells. Use the blank to set the instrument to zero absorbance and determine the absorbances of standards and samples. Prepare a calibration curve of absorbance vs. concentration and determine the chloride concentration in the samples. The calibration correlation coefficient shall be ≥ 0.999.

B.2.2.2 Calculation. Calculate the chloride content of the hydrazine using the formula in B.2.1.3.

B.2.2.3 Reagents and equipment. The following shall apply as test conditions of B.2.2:

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B.2.2.3.1 Reagents.

a. Purified water of 18 M-ohm or greater resistivity.

b. Ferric alum indicator, Volhard (oxidation reduction indicator) – Dissolve 10 g of ferric ammonium sulfate dodecahydrate ($FeNH_4(SO_4)_2 \cdot 12H_2O$) ACS reagent grade in ~ 50 mL of purified water. Add 10 mL of ACS reagent grade nitric acid (HNO_3), and dilute to 100 mL with purified water. Ferric alum indicator solution may be commercially obtained.

c. Mercuric thiocyanate, 4.17 g/L in methanol for chloride determination (commercially available).

d. Chloride standard, 1000 mg/L (may be commercially obtained; must be certified).

B.2.2.3.2 Equipment. UV-Vis Spectrophotometer, double beam, equipped with 5 cm cells.

B.2.3 Ion chromatography (IC) method.

B.2.3.1 Procedure. To the hydrazine NVR in one vessel from 4.3.6, add an appropriate amount of purified water (18 M-ohm or greater resistivity) such that the instrument reading is within the instrument calibration curve. Gently warm the mixture for 3 to 5 minutes (min) ensuring the water contacts the surface area of the vessel that the fuel mixture came in contact with. Allow the solution to cool. Transfer a representative aliquot into an appropriate labeled IC vial and analyze.

B.2.3.2 Calibration. Prepare a series of calibration standards in purified water to bracket the expected concentration range of the samples by combining appropriate aliquots of a certified Cl^- or multi-element IC anion standard solution. Prepare and/or commercially obtain a mid-range check standard from a different stock standard source to verify the calibration (Initial Calibration Verification, ICV). Prepare a calibration curve by plotting the instrument response (integrated peak area) vs. the Cl^- standard concentration to determine the sample concentration. Alternatively, use the quantitation method recommended by the instrument manufacturer.

B.2.3.3 Quality control.

(1) The calibration correlation coefficient must be ≥ 0.995.

(2) Analyze the ICV every 10 samples to validate the instrument calibration. The ICV should fall within ± 10% of its concentration.

B.2.3.4 Calculation. Calculate the chloride content of the hydrazine using the formula in B.2.1.3. If the dilution factor is accounted for by the instrument adjust the formula accordingly.

B.2.3.5 Reagents.

a. Purified water of 18 M-ohm or greater resistivity.

b. Eluent mix stock, approximately equivalent to 3.5 mM sodium carbonate/1.0 mM sodium bicarbonate – If the IC system is equipped with an automatic eluent generator, follow the instrument owner's manual instructions. Otherwise, combine an appropriate aliquot of a commercially available IC sodium carbonate/sodium bicarbonate concentrate in a volumetric flask containing purified water, fill to the mark, and mix thoroughly. If preparing the mix in-house dissolve 370.9608 mg ACS reagent grade sodium carbonate and 84.007 mg sodium bicarbonate in a 1 L volumetric flask, and proceed as above. Eluent generator systems may use alternate eluents as long as they are paired with a column that can effectively quantitate 1 ppmw Cl^-.

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c. Chloride or multi-element IC anion standard solution (must be certified).

B.2.3.6 Equipment. The ion chromatograph must have the following components:

(1) Eluent pump capable of delivering 1 - 5 mL/min of eluent at a nominal flow and pressure specified by the instrument manufacturer.

(2) Sample pump capable of filling a sample loop with no resistance.

(3) Sample loop capable of delivering 5 to 100 µl of sample to the analytical column.

(4) An ion exchange column suitable for resolving chloride.

(5) A suppressor device.

(6) Conductivity cell.

(7) Data acquisition device.

(8) Five decimal place analytical balance (required only if eluent mix is prepared in-house).

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APPENDIX C

IRON

C.1 SCOPE

C.1.1 Scope. This Appendix describes three methods for the determination of iron content of the hydrazine. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

C.2 METHODS

C.2.1 Inductively-coupled argon plasma atomic emission spectrometry. For each instrument interference effects must be evaluated, corrected as applicable (ASTM D1976), and instrument performance validated. The argon gas supply must have a purity of 99.99% or better. Choose the iron wavelength recommended by the instrument manufacturer or that which provides the highest sensitivity (most commonly used is 259.940 nm).

C.2.1.1 Procedure. Rinse the hydrazine NVR contained in one vessel from 4.3.6 as follows: Add 5 mL of 5N HNO_3 to the vessel and place on a hot plate. Heat the mixture to near boiling, maneuver beakers so the acid contacts the surface area of the glass that the fuel mixture came in contact with, and maintain at that temperature for three to five minutes. Allow the solution to cool and transfer to a 100 mL volumetric flask. Rinse the vessel at least three times with 5 mL aliquots of 0.1N HNO_3 combining all rinses in the volumetric flask. Dilute to the mark with purified water and mix thoroughly.

NOTE: Sulfuric acid may be substituted for nitric acid (nitric acid preferred) in C.2.1.1 provided standards and samples are matrix matched (applies to all of C.2).

C.2.1.2 Calibration. Prepare an acid matrix blank and an Fe standard in a 0.1N HNO_3 acidified purified water solution to bracket the expected concentration range of the samples by combining an appropriate aliquot of a certified Fe stock standard. Prepare a mid-range check standard, from a different stock standard source to verify the calibration. Determine the concentration from the calibration curve prepared from the Fe standard using the acid matrix blank as the zero point, and record the Fe concentration of the sample.

C.2.1.3 Quality control. The instrumental drift shall be checked using the acid matrix blank and standard at the beginning of each run, at least every ten readings during the run, and at the end of the run. The blank shall read ≤ the detection limit and the standard shall fall within ± 10% of the standard value. If the recovery of the analyte falls outside of the required control limits, the source of the problem must be identified and resolved before continuing analyses.

C.2.1.4 Calculation. Calculate the iron content of the hydrazine using the following formula:

$$Fe\ ppmw = \frac{(Fe_{ppmv})(V_d)}{(V_p)(\rho)}$$

where,

Fe_{ppmv} = Iron content of the sample solution from the calibration curve.

V_d = Final dilution volume taken from NVR in mL.

V_p = Volume of hydrazine used in the NVR analysis in mL.

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ρ = Density of hydrazine (1.01 g/mL).

C.2.1.5 Reagents and equipment. The following shall apply as test conditions of C.2.1.

C.2.1.5.1 Reagents.

a. Purified water of 18 M-ohm or greater resistivity.

b. Iron standards, certified commercially obtained solutions.

c. Nitric or sulfuric acid, concentrated, suitable for trace metal analysis.

C.2.1.5.2 Equipment.

a. Hot plate – controllable heat.

b. Inductively-coupled argon plasma atomic emission spectrometer (ICP-AES).

C.2.2 Atomic absorption.

C.2.2.1 Procedure. Prepare the sample solution in the same manner as described in C.2.1.1 or dilute the hydrazine sample 1:2 and use directly. Determine the absorbance of the sample at an Fe wavelength closest to 2480 Å on a suitable atomic absorption spectrophotometer, after setting the instrument to zero absorbance, with 0.1N HNO_3. Compare the results to a calibration curve prepared from the Fe standards and record the Fe concentration of the sample.

C.2.2.2 Calibration. Prepare a minimum of three calibration standards to bracket the expected Fe concentration range of the samples and a mid-range check standard to verify the calibration in the same manner as described in C.2.1.2. After setting the instrument to zero absorbance with 0.1N HNO_3, determine the absorbance of the calibration standards and check standard. Prepare a calibration curve of absorbance vs. Fe concentration.

C.2.2.3 Calculations. Determine the iron content of the hydrazine as follows,

C.2.2.3.1 Iron from the NVR. Use the formula in C.2.1.4.

C.2.1.3.2 Iron from direct hydrazine analysis.

$$Fe\ ppmw = \frac{(Fe_{ppmv})(DF)}{\rho}$$

where,

DF = Dilution factor used to dilute hydrazine.

ρ = Density of hydrazine (1.01 g/mL).

C.2.2.4 Quality control. Follow the guidelines described in ASTM D1068 Method A with the following exception:

a. The mid-range standard from C.2.1.2 shall be used as the laboratory control sample.

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C.2.2.5 Reagents and equipment. The following equipment shall apply as test conditions of C.2.2.

C.2.2.5.1 Reagents. Same as in C.2.1.5.1.

C.2.2.5.2 Equipment. Atomic absorption spectrophotometer with an iron hollow cathode lamp.

C.2.3 Colorimetric method.

C.2.3 1 Procedure. ASTM E394 shall be followed with the following exceptions:

a. Add 5 mL of the ammonium acetate-acetic acid buffer solution to each empty 100 mL volumetric flasks (for standards and samples) and set aside.

b. Dissolve the residue of one vessel from 4.3.6 by adding 1 mL of concentrated acid (HNO_3 or H_2SO_4 as applicable) and carefully adding 50 mL purified water. Mix thoroughly.

c. Add 1 mL of concentrated acid (HNO_3 or H_2SO_4 as applicable), and 50 mL purified water to at least four empty 150 mL evaporating dishes for the standards.

NOTE: Make sure solution is thoroughly mixed after each component addition.

d. Add the 2 mL of hydroxylamine hydrochloride and the 5 mL of 1,10-phenanthroline as described in the method.

e. Add enough sodium acetate solution to bring the pH of the mixture to between pH 3 and pH 4 while swirling (add ~ 20 mL initially and then continue adding drop wise until desired pH is reached).

f. Transfer the solutions to their respective 100 mL volumetric flask from step a above, dilute to the mark, and mix thoroughly. Allow the flasks to set at room temperature in the dark for a minimum of 15 min.

g. Determine the Fe concentrations from the calibration curve prepared from the Fe standards; the correlation coefficient shall be ≥ 0.990.

C.2.3.2 Calculation. Calculate the iron content of the hydrazine using the formula in C.2.1.4.

C.2.3.3 Reagents. The following reagents are in addition to those reagents listed in ASTM E394,

a. Sodium acetate solution. Dissolve 200 g $CH_3COONa•3H_2O$ in 1 L volumetric flask with 700 mL of purified water and dilute to the mark. An equivalent solution may be obtained commercially or prepared from anhydrous sodium acetate.

b. Nitric or sulfuric acid, concentrated, suitable for trace metal analysis.

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NON-VOLATILE RESIDUE

D.1 SCOPE

D.1.1 Scope. This Appendix describes three methods for the determination of non-volatile residue of the hydrazine. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

D.2 METHODS

D.2.1 Hot plate. This method involves the gradual evaporation of a hydrazine sample on an explosion proof hot plate in a fume hood. The difference in weight before and after evaporation is calculated as NVR.

D.2.1.1 Procedure. Place three clean borosilicate glass or platinum evaporating dishes (latter preferred) in an oven at 110 °C for 30 min. Label one as a blank and the others as samples A1 and A2. Remove evaporating dishes from the oven, place in a desiccator to cool for 15 min, and then place next to a five place balance long enough to equilibrate (usually around 30 min). Weigh each clean, dry evaporating dish to the nearest 0.0001 g until the weight remains constant to ± 0.1 mg. Record the sample evaporating dish weight as A1, A2 and the blank evaporating dish weight as C. Add 40 mL of water to each one of the evaporating dishes. Transfer 40 mL of hydrazine sample into the evaporating dishes marked A1 and A2. Hydrazine and water form an azeotrope (constant boiling mixture) with a maximum boiling point of 120.5 °C (771 mmHg pressure). Therefore, when anhydrous hydrazine and water are mixed, a corresponding eutectic of 76.0 mole percent hydrazine is the maximum concentration obtained. This eliminates any significant hazard in evaporating a hydrated sample of hydrazine to dryness. Place each evaporating dish on a hot plate which is set to a temperature of 150 – 200 °C. Evaporate to dryness, but do not bake. Remove the evaporating dishes from the explosion proof hot plate and place them in oven set at 110 °C for 30 min. Remove dishes from oven and place in a desiccator for 15 min to cool; then place them next to a five place balance long enough to equilibrate. Weigh each evaporating dish as before to a constant weight. (NOTE: A/C fluctuations/cycling may significantly affect the weights obtained for the evaporating dishes; a narrow range viscosity thermometer is recommended for monitoring room temperature during the weighing process). Record the sample evaporating dish weight as B1, B2 and the blank evaporating weight as D.

NOTE

Save the propellant residue/hydrazine NVR for chloride (4.3.4) and iron (4.3.5) determinations.

D.2.1.2 Calculation. Calculate the non-volatile residue of the hydrazine as follows:

$$NVR\ mg/L = \left[\frac{(Wt_B - Wt_A) - (Wt_D - Wt_C)}{Sample\ volume_{mL}}\right] \times 10^6$$

where,

A = Weight of the sample evaporating dish, g.

B = Weight of the sample evaporating dish and residue, g.

C = Weight of blank evaporating dish, g.

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D = Weight of blank and water residue, g.

D.2.1.3 Equipment.

a. Hot plate, explosion proof.

b. High efficiency fume hood.

c. Five decimal place analytical balance (semi-micro balance).

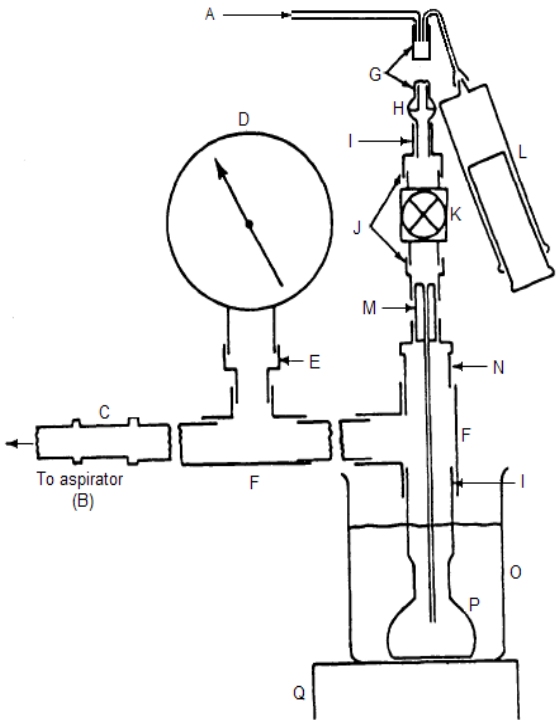
D.2.2 Dropwise method.

D.2.2.1 Procedure. Clean the unassembled apparatus with detergent and rinse thoroughly with purified water. Dry the parts in an oven at 110 °C. Assemble the apparatus as shown in Figure D-1. Fill the water bath with purified water and turn on the heater. Adjust the water level of the bath to approximately one-half inch below the lowest fitting and mark this level on the bath. When the temperature bath reaches boiling, adjust the N_2 purge to ~ 2 L/min and turn on the aspirator. With the sample delivery valve closed, the pressure in the apparatus should be below 30 mmHg. Open the sample delivery valve slowly to adjust the apparatus pressure to ~ 60 mmHg. Allow the apparatus to purge for ~ 5 min to remove traces of water. Remove the water bath and dry the outside of the residue vessel with a clean lintless tissue. Disconnect the aspirator from the apparatus and remove the residue vessel using a tissue to hold the vessel. Wipe the outside of the vessel and place it near the analytical balance. After at least 20 min, accurately weigh the residue vessel until the weight remains constant to ± 0.1 mg. Record the weight. Reassemble the apparatus and close the sample delivery valve. Replace the water bath, adjust the water level to the mark, and heat the water to boiling. Transfer 100 mL of hydrazine to the apparatus. With the N_2 purge in place, turn on the water aspirator and reduce the apparatus pressure to less than 30 mmHg. Slowly open the sample delivery valve to admit hydrazine to the residue vessel and adjust the flow such that the apparatus pressure does not exceed 60 mmHg (~ 2 drops per second). The quantity of hydrazine in the residue vessel should not exceed 2 - 4 mL at the equilibrium evaporation rate. Adjust the sample delivery valve to maintain the evaporation pressure at less than 60 mmHg. After 100 mL of hydrazine have been evaporated (~ 30 min), allow the residue vessel to purge for an additional 5 min to ensure complete evaporation of volatile material. Remove the water bath, disconnect the aspirator, wipe the outside of the residue vessel, and remove it from the apparatus. Allow the residue vessel to stand near the analytical balance for at least 20 min before weighing. Weigh the residue vessel as above, and record the weight. To clean the residue vessel reflux 5 mL of 5N ACS reagent grade H_2SO_4 (or equivalent) in the vessel for several minutes followed by several rinses of purified water. The vessel may be dried as described above or by gently heating the vessel with a burner while purging with dry N_2. Regardless of the drying method used, allow at least 20 min for the vessel to equilibrate to ambient temperature and humidity before weighing.

NOTE

Save the propellant residue/hydrazine NVR for chloride (4.3.4) and iron (4.3.5) determinations.

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A - GN_2

B - Water aspirator (glass or polyethylene)

C - Connector (polyethylene, separable)

D - Gauge (stainless steel, absolute pressure, 0 - 760 mmHg range)

E - Adapter (Swagelok SS-8-TA-7-4 or equivalent)

F - Union Tee (Swagelok SS-810-3 or equivalent; 2 ea.)

G - Pipette (100 mL, volumetric, borosilicate glass)

H - Joints (borosilicate glass, spherical, 12/5, ball, socket and clamp)

I - PTFE Ferrules (Swagelok T-814-1, T-813-1, T-404-1, T-403-1, or equivalent)

J - Connector (Swagelok SS-400-7-2 or equivalent)

K - Sample delivery valve (Swagelok SS-ORM2 or equivalent)

L - Syringe (100 mL glass with an 8 – 12 in PTFE needle).

M - Rod (PTFE or polyethylene, $1/4$ in OD & a $1/16$ in OD x 0.025 in wall tube inserted)

N - Reducer (Swagelok SS-400-R-8 or equivalent)

O - Beaker (glass, 1000 mL capacity)

P - Flask (25 mL volumetric, borosilicate glass)

Q - Hot plate (adjustable heat range)

NOTE: All fittings, stainless steel, Swagelok or equivalent.

FIGURE D-1. *NVR apparatus.*

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D.2.2.2 Calculation. Calculate the non-volatile residue of the hydrazine as follows:

$$NVR\ mg/L = \left[\frac{W_2 - W_1}{V}\right] \times 10^6$$

where,

W_1 = Weight of clean vessel, g.

W_2 = Weight of residue and vessel, g.

V = Volume of sample used, mL.

Report the averaged result of duplicate NVR determinations.

D.2.2.3 Equipment. The following equipment shall apply as test conditions of D.2.2.

a. Components described in Figure D-1.

b. Tubing – polyethylene, ½ in OD, borosilicate glass, ½ in OD.

c. Clamps – hose, to fit ½ in OD tube.

d. Five decimal place analytical balance (semi-micro balance).

D.2.3 Rotary evaporation.

D.2.3.1 Procedure. Dry a 250 mL round bottom flask in an oven at 110 °C for one hour. Allow the flask to cool in a desiccator and weigh on the analytical balance until the weight remains constant to ± 0.1 mg. Transfer 100 mL of hydrazine into the 250 mL flask. Place the flask on the evaporator using a standard taper Teflon sleeve as a seal. Attach a water aspirator pump to the evaporator via a one liter suction flask and evaporate the hydrazine to dryness. A water bath may be placed around the rotating flask to aid evaporation.

WARNING

The contact of hydrazine vapor with metal surfaces, as in a metal rotary evaporator, must be avoided. Do not exceed 40 °C during the initial evaporation phase when hydrazine liquid is present. Chemical splash proof goggles must be worn throughout the entire evaporation process.

When no more hydrazine evaporates, the water bath is lowered. Final high boiling components (mainly aniline) are distilled under vacuum at temperatures up to 100 °C using a hot air gun for heating. The flask is then removed and dried in an oven at 110 °C for one hour and weighed as before to a constant weight.

NOTE

Save the propellant residue/hydrazine NVR for chloride (4.3.4) and iron (4.3.5) determinations.

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D.2.3.2 Calculation. Use equation in D.2.2.2.

D.2.3.3 Equipment.

a. All glass rotary evaporator, BÜCHI Rotavapor or equivalent.

b. Flask, round bottom, 250 mL, with standard neck to fit evaporator.

c. Water aspirator – glass or polyethylene.

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ANILINE

E.1 SCOPE

E.1.1 Scope. This Appendix describes three methods for the determination of aniline content of the hydrazine. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

E.2 METHODS

E.2.1 Spectrophotometric method.

E.2.1.1 Procedure. For high purity, pipette or syringe 5 mL of hydrazine into a 25 mL volumetric containing ~ 15 mL of purified water; for monopropellant pipette or syringe 0.10 mL (100 µL) of hydrazine. Dilute to the mark with purified water, and mix the solution thoroughly. Fill two 1 cm pathlength silica spectrophotometer cells with purified water and place one in each position for sample and reference cells. Using the appropriate source and detector for the spectrophotometer, set the absorbance to zero at 320 nm and scan the region between 320 and 240 nm. If the baseline is relatively stable, the cells and purified water can be considered free from interfering substances. Exchange the water in the sample cell with the previously prepared sample, using several rinses with the sample solution, and rescan the 320 to 240 nm region, first ensuring that the instrument is set to zero absorbance at 320 nm. Using a baseline under the aniline peak at 280 nm, measure, and record the net absorbance of the peak. Determine the aniline content of the sample by comparing the net absorbance obtained to a calibration curve of net absorbance vs. percent aniline.

E.2.1.2 Calibration. Pipette or syringe 0.025 mL (25 µL) of ACS reagent grade aniline into a 100 mL volumetric flask containing 50 mL of purified water. Shake the mixture to dissolve the aniline and dilute the solution to the mark with purified water. Mix the standard thoroughly. This standard is equivalent to 0.025 wt%. Prepare a series of calibration standards in the aniline range expected to find in the samples. Determine the net absorbance of standards vs. purified water in the same manner as with the sample. Plot net absorbance vs. percent aniline and draw a curve through the points. A check standard shall be prepared fresh daily to check the calibration standards. If the check standard falls outside of ± 5% of the expected value, fresh calibration standards shall be prepared and the calibration check repeated until the required accuracy is demonstrated.

E.2.1.3 Equipment. Spectrophotometer, double beam, ultra violet, narrow band width, recording, equipped with 1 cm silica cells.

E.2.2 Gas chromatographic method – Stabilwax®-DB (carbonaceous material determination).

E.2.2.2 Procedure. Follow the procedure described in G.2.1.1.

E.2.2.3 Calibration. Follow the calibration procedure as described in G.2.1.2. Add ~ 120 µL ACS reagent grade aniline to the OVCM stock standard solution, and prepare the working standard as described in G.2.1.2.2.

E.2.2.4 Calculation. Calculate the aniline content of the hydrazine using the equations in G.2.1.3.

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E.2.3 Gas Chromatographic method - Carbowax 20M.

E.2.3.1 Procedure. A 0.53 mm OD x 15 m crosslinked “Carbowax 20M” capillary column shall be used. Helium is used as a carrier gas. A gradient temperature program is used from 40 °C to 130 °C. Splitless injection is used. Sample size should be 0.4 to 1 µL. Condition the column with several injections of the calibration mixture three minutes apart. After the fifth injection, allow sufficient time for the final injection of aniline to elute, ~ 10 min. Adjust the flow rate to allow for baseline separation of the trace monomethylhydrazine from hydrazine.

E.2.3.2 Calibration (high purity hydrazine). Pipette or syringe 0.5 mL of aniline into a 1 L volumetric flask containing ~ 500 mL of purified water, and mix the solution thoroughly. Fill to the mark with purified water and thoroughly mix. Pipette 10 mL of this solution into a 100 mL volumetric flask and dilute to the mark. This calibration standard is equivalent to 0.005 wt% aniline in hydrazine. Inject 1 µL of the standard into the inlet of the chromatograph and record the area of the aniline peak.

E.2.3.3 Calculation. Use the following formula to calculate the aniline content of the hydrazine:

$$Aniline\ wt\% = \frac{0.005\ A_{Sample}}{A_{Std}}$$

where,

A_{Sample} = Area of the sample aniline peak multiplied by its attenuation factor.

A_{Std} = Area of the standard aniline peak multiplied by its attenuation factor.

E.2.3.4 Equipment. The following equipment shall apply to test conditions of E.2.3.

a. Gas chromatograph, equipped with a flame ionization detector (FID).

b. Agilent ChemStation or equivalent.

c. A 0.53 mm OD x 15 m crosslinked “Carbowax 20M” capillary column.

E.2.4 Alternate capillary method.

E.2.4.1 Procedure for monopropellant grade hydrazine. Aniline content may also be determined using the assay capillary column method described in Appendix A of this specification (A.2.1).

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CARBON DIOXIDE

F.1 SCOPE

F.1.1 Scope. This Appendix describes a method for the determination of carbon dioxide content of the hydrazine. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance. Determination of the CO_2 content in the hydrazine must be performed before all other analyses to prevent further absorption of CO_2.

F.2 METHOD

F.2.1 Determination of carbon dioxide by non-dispersive IR analysis. This method is to be utilized to determine the CO_2 concentration in hydrazine within a 10 - 100 mg/kg (ppmw) range. The sample of hydrazine is injected into a solution of 30% phosphoric acid which absorbs the hydrazine and ammonia. Carbon dioxide is liberated from the acidic solution and is swept out of the solution by a stream of N_2 gas through a silica gel drying tube to remove water vapor, and is measured by an infrared technique.

F.2.1.1 Calibration procedure.

a. Assemble the apparatus as shown in Figure F-1.

b. Adjust the flow of N_2 purge gas according to the manufacturer's recommendation by use of the control valve and flow meter.

c. Add an appropriate amount of 30% phosphoric acid containing several drops of bromocresol green or methyl red indicator to the CO_2 reaction flask, turn on the N_2 gas to give vigorous, constant mixing. The N_2 gas flow (~ 200 mL/min) may need adjustment to obtain reproducible results but must remain constant throughout the analysis.

d Allow the instrument to stabilize and adjust the zero.

F.2.1.2 Standardization.

If using direct instrument response comparison (flow technique),

a. Inject via the septum on the phosphoric acid containing unit, an appropriate amount of a prepared 100 ng/µL (ppmv) working CO_2 carbonate or bicarbonate standard into the phosphoric acid solution to achieve a reading of 100 ppm on the instrument (~ 200 mL). Repeat the injection to ensure reproducibility is within ± 10 ppm and record.

b. Once reproducibility is verified, repeat as above with ~ 100 µL (50 ppm) and 50 µL (25 ppm) and check for linearity.

If using a data acquisition device,

a. Prepare a 5 point calibration curve using the 100 ppmv working standard to bracket the expected concentration range of the samples using a gas-tight syringe (10 µL is equivalent to 1000 ng).

b. Plot the absorbance counts of the standards vs. nanograms. to determine the CO_2 concentration in the samples. The calibration correlation coefficient shall be ≥ 0.995.

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F.2.1.3 Sample analysis. Inject an appropriate amount of hydrazine sample into the phosphoric acid reaction flask to obtain a reproducible result within the standards range. Record the reading from the infrared analyzer scale or data acquisition device (from the zero point). Perform the sample analysis in triplicates (verifying reproducibility) and use the average to determine the CO_2 concentration.

F.2.1.3.1 Calculations. Determine the carbon dioxide content of the hydrazine as follows,

a. Flow technique,

$$CO_2\ Wt\% = \frac{(Std\ conc_{ppmv})(Std\ vol_{\mu L})(Sample\ response)}{(Sample\ vol_{\mu L})(Std\ response)(\rho)(1 \times 10^4)}$$

where,

ρ = Density of hydrazine (1.01 g/mL).

b. Data acquisition device,

$$CO_2\ Wt\% = \frac{Sample\ CO_{2\ ng}}{(Sample\ vol_{\mu L})\ (\rho)(1 \times 10^4)}$$

where,

ρ = Density of hydrazine (1.01 g/mL).

F.2.1.4 Reagents.

a. Purified water, CO_2 free - Prepare by boiling purified water vigorously for 15 min or by bubbling N_2 gas through it for several hours.

b. Carbon Dioxide Stock Solution, 1000 ppm - Dissolve 2.4083 g ACS reagent grade sodium carbonate or 1.9088 g ACS reagent grade sodium bicarbonate, previously dried for one hour at 100 °C, in CO_2 free purified water in a 1 L volumetric flask; fill to the mark and mix thoroughly. Transfer 1 mL of this solution to a 10 mL volumetric flask, fill to the mark with CO_2 free purified water to prepare a 100 ppm standard. Prepare fresh daily.

c. Phosphoric Acid, 30% by volume - Dilute 355 mL of 85% H_3PO_4 to 1 L with purified water. Add enough indicator to this solution to readily determine when the acid solution is depleted.

d. Dry N_2 gas supply.

e. Silica gel desiccant.

F.2.1.5 Equipment.

a. Non-dispersive infrared analyzer, Rosemount Analytical Model XStream-X2GP or equivalent.

b. Glass reaction tube with replaceable septum.

c. PTFE tubing.

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d. Glass drying tube (should not be larger in outer diameter than the PTFE tubing used to transport the sample to the detector).

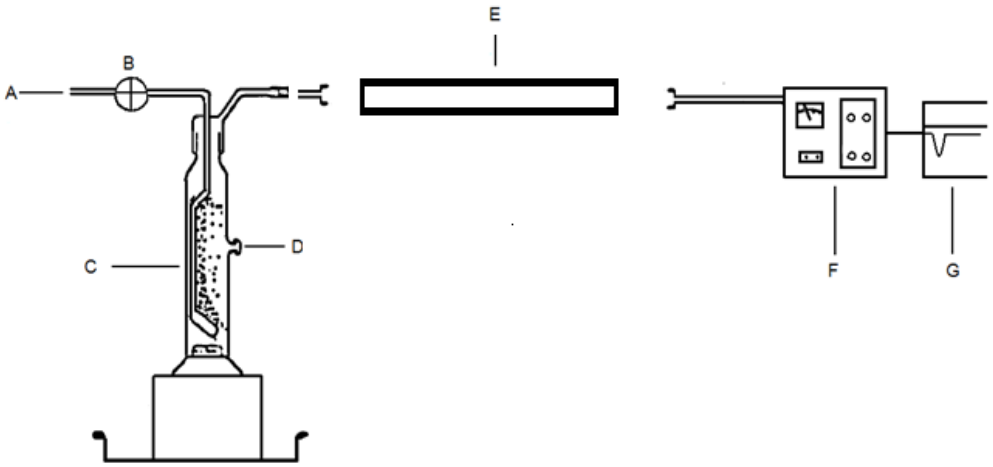
e. Needle valve to control flow into reaction tube.

f. Fittings to achieve air tight connections.

g. Operating fume hood.

h. Gas-tight syringe.

i. Suitable data acquisition device.



A	-	Sparge gas inlet	E	-	Trap for removing H_2O
B	-	Needle valve	F	-	Infrared analyzer
C	-	Reactor and phosphoric acid solution	G	-	Recorder
D	-	Septum inlet for hydrazine		-	

FIGURE F-1. *CO_2 apparatus.*

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APPENDIX G

CARBONACEOUS MATERIAL

G.1 SCOPE

G.1.1 Scope. This Appendix describes a method for the determination of the carbonaceous material (other volatile carbonaceous material (OVCM) + other carbonaceous material (OCM)) content of the hydrazine. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

G.2 METHOD

G.2.1 Gas chromatographic method.

G.2.1.1 Procedure. A 60 m x 0.53 mm x 1.00 µm film thickness Stabilwax®-DB column, an FID equipped with a split injector, and an ultra-inert, deactivated, glass wool packed split liner shall be used. Table G-1 provides a list of the GC parameters for initial setup. These parameters may vary depending on GC system and detector brands, and may be adjusted to provide adequate component resolution.

TABLE G-1. *GC parameters.*

PARAMETER	CONDITION
Injector Port Temp	175 °C
Injection Mode	Split injection
Sample Volume.	1 µL
Split Ratio	2:1
Carrier Gas	He
Control Flow	10 mL/min
Control Mode	Flow
FID Detector Temp.	210 °C
Equilibration Time	2.00 min
Initial Temp.(T_0)	60 °C
Initial Time (t_0)	3.00 min
Rate (1)	10 °C/min
Final Temp. (T_{f1})	100 °C
Final Time (t_{f1})	0.00 min
Rate (2)	20 °C/min
Final Temp. (T_{f2})	185 °C
Final Time (t_{f2})	3.75 min

The elution order is as follows:

UDMH < IPA < MMH < hydrazine < aniline

G.2.1.2 Calibration. Prepare a standard stock solution, and a working standard of UDMH, IPA, and MMH in hydrazine (N_2H_4) as described in G.2.1.2.1 and G.2.1.2.2 (deviations are allowed as long as the working standard is conducive to the analysis). Run a blank of high purity N_2H_4 meeting this specification and subtract the areas of UDMH, IPA, and MMH from the areas of the standard. The signal

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to noise ratio shall be such that it allows for the effective quantitation of at least 5 ppmw MMH. If any of the contaminants are found in a N_2H_4 sample, analyze the standard and sample(s) in triplicate; the RSD ≤ 10%.

G.2.1.2.1 Standard stock solution.

1. Weigh a 40 mL volatile organic analysis (VOA) type septa vial with its cap to the nearest 0.0001 g and record the weight.

2. Add ~ 30 mL of high purity N_2H_4 meeting this specification to the vial; cap, weigh to the nearest 0.0001 g, and record the weight. The difference in weight before and after the addition is the amount of N_2H_4 added.

3. Using a gas-tight glass syringe add ~ 190 µL UDMH to the vial, weigh to the nearest 0.0001 g and record the weight. Subtract the weight of the high purity N_2H_4 from the weight obtained to determine the weight of UDMH added.

4. Follow same procedure as above with the additions of ~ 190 µL IPA, 170 µL MMH, and 120 µL aniline if applicable.

5. Mix the solution thoroughly and store tightly sealed in the fuel hood under a N_2 gas purge.

G.2.1.2.2 Working standard.

1. Weigh a suitable size VOA-type vial with its cap to the nearest 0.0001 g. Record the weight.

2. Add an appropriate amount of high purity N_2H_4 to the vial, cap, and weigh following the same procedure as in G.2.1.2.1. The difference in weight before and after the addition is the amount of N_2H_4 added.

3. Shake the standard stock solution and inject enough into the vial to obtain a mid-range concentration based on the maximum allowed limits for high purity and monopropellant N_2H_4 and weigh to the nearest 0.0001 g. Subtract the weight of the high purity N_2H_4 from the weight obtained to determine the weight of standard stock solution added.

G.2.1.3 Calculations.

a. Standard stock solution:

$$Analyte\ ppmw_{Stock\ Std} = \frac{Wt_{Analyte} \times 10^6}{Wt_{Total}}$$

where,

$$Wt_{Total} = Wt_{N_2H_4} + Wt_{UDMH} + Wt_{IPA} + Wt_{MMH} + Wt_{aniline}$$

b. Working standard:

$$Analyte\ ppmw_{Wk\ Std} = \frac{Analyte\ ppmw_{Stock\ Std} \times Wt_{\ Stock\ Std\ added}}{Wt_{Stock\ Std\ added} + Wt_{N_2H_4}}$$

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c. Sample calculation:

$$Analyte\ ppmw_{Sample} = \frac{Analyte\ ppmw_{Wk\ Std} \times A_s}{A_{analyte}}$$

where,

$A_{analyte}$ = The area of the analyte peak times its signal attenuation factor.

A_s = The area of the respective analyte in the sample times its signal attenuation factor.

Calculate OVCM in weight percent using the following equation,

$$OVCM\ Wt\% = UDMH\ Wt\% + IPA\ Wt\% + MMH\ Wt\%$$

Calculate OCM in weight percent by addition of all impurities providing a positive FID response ahead of and trailing the hydrazine peak excluding OVCM and aniline. Calculate the content in the N_2H_4 sample based on the MMH factor as follows,

$$OCM\ wt\% = \left[\frac{MMH\ ppmw_{Wk\ Std} \times A_i}{A_{MMH}}\right] \times 10^{-4}$$

where,

A_i = The average (based on triplicates) of the sum of all the peak areas excluding OVCM and aniline.

A_{MMH} = The average peak area obtained for the MMH standard.

Report the carbonaceous material content of the N_2H_4 as the sum of OVCM + OCM.

G.2.1.4 Reagents and equipment. The following reagents and equipment shall apply as test conditions of G.2.1.

G.2.1.4.1 Reagents.

a. N_2H_4 - High purity grade conforming to this specification.

b. UDMH - Conforming to MIL-PRF-25604.

c. IPA - ACS reagent grade.

d. MMH - Conforming to MIL-PRF-27404.

e. Aniline - ACS reagent grade.

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G.2.1.4.2 Equipment.

(1) Gas chromatograph equipped with an FID and a split injector.

(2) Agilent ChemStation or equivalent.

(3) A 60 m x 0.53 mm ID and 1.00 µm film thickness Stabilwax®-DB column.

(4) Ultra-inert deactivated liner, split, straight, glass wool.

(5) No-bleed septa.

(6) Gas-tight syringe.

(7) Analytical balance calibrated to 0.0001 g.

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CONCLUDING MATERIAL

Custodians
Navy – AS
Air Force – 68
DLA – PS

Preparing Activity
Air Force – 68
(Project: 9135-2021-004)

Review Activities
Air Force - 19

Civil Agency
NASA - NA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information using the ASSIST Online database at https://assist.dla.mil.

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